REAL ANALYSIS

KEIGO TAMAKI

2025

1. RIEMANN INTEGRATION

1. DEFINITION OF THE RIEMANN-STIELTJES INTEGRAL

Definition 1.1 (Partition)**.** Let $[a, b]$ be a closed interval for some $a, b \in \mathbb{R}$. A *partition* P of $[a, b]$ is a finite set of points $\{x_0, x_1, \ldots, x_n\}$ where

$$a = x_0 \leq x_1 \leq \cdots \leq x_{n-1} \leq x_n = b$$

and we write $\Delta x_i = x_i - x_{i-1}$ for $i \in \{1, \ldots, n\}$.

Using partitions, we first construct the *Riemann integral*. One may have learned this approach in an earlier calculus course, but we construct it in a more rigorous manner.

Definition 1.2 (Riemann Integration)**.** Let $[a, b]$ be a closed interval for some $a, b \in \mathbb{R}$, and let P be a partition of $[a, b]$. Let f be a real valued function that is bounded on $[a, b]$.

Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. For each $i \in \{1, \ldots, n\}$, denote

$$M_i = \sup_{[x_{i-1}, x_i]} f, m_i = \inf_{[x_{i-1}, x_i]} f$$

and denote

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i, L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i.$$

The *upper Riemann Integral* is defined by

$$\int_{-a}^{b} f(x)dx = \inf_P U(P, f)$$

and the *lower Riemann Integral* is defined by

$$\int_{a}^{-b} f(x)dx = \sup_P L(P, f).$$

We say that f is *Riemann integrable* if $\int_{-a}^{b} f(x)dx = \int_{a}^{-b} f(x)dx$ (i.e. the upper and lower Riemann integrals are equal), and we denote $f \in \mathcal{R}$, the set of all Riemann integrable functions. If f is Riemann integrable, then the *Riemann integral* of f is defined by

$$\int_{a}^{b} f(x)dx = \int_{-a}^{b} f(x)dx = \int_{a}^{-b} f(x)dx$$

Next, we define the Riemann-Stieltjes integral, the integral of main interest within this chapter. The Riemann-Stieltjes integral shares many similarities with the Riemann integral, but with a significant improvement.

Notice that in the Riemann integral, each "box" has a width equivalent to the length of a chosen interval. Essentially, the Riemann-Stieltjes integral allows more freedom in deciding the width of each "box", where instead of using simply the length of a chosen interval, we apply a separate function to the endpoints of each interval and define the width as the difference between the respective outputs.

Definition 1.3 (Riemann-Stieltjes integral)**.** Let $[a, b]$ be a closed interval for some $a, b \in \mathbb{R}$, and let P be a partition of $[a, b]$. Let f be a real valued function that is bounded on $[a, b]$, and let α be a monotonically increasing real valued function that is bounded on $[a, b]$.

Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. For each $i \in \{1, \ldots, n\}$, denote $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$, and similar to earlier, denote $M_i = \sup_{[x_{i-1}, x_i]} f, m_i = \inf_{[x_{i-1}, x_i]} f$, and

$$U(P, f, \alpha) = \sum_{i=1}^{n} M_i \Delta\alpha_i, L(P, f, \alpha) = \sum_{i=1}^{n} m_i \Delta\alpha_i.$$

The *upper Riemann-Stieltjes Integral* is defined by

$$\int_{-a}^{b} f d\alpha = \inf_P U(P, f, \alpha)$$

and the *lower Riemann-Stieltjes Integral* is defined by

$$\int_{a}^{-b} f d\alpha = \sup_P L(P, f, \alpha).$$

We say that f is *Riemann integrable with respect to* α or *Riemann-Stieltjes integrable* if $\int_{-a}^{b} f d\alpha = \int_{a}^{-b} f d\alpha$ (i.e. the upper and lower Riemann-Stieltjes integrals are equal), and we denote $f \in \mathcal{R}(\alpha)$, the set of all Riemann integrable functions with respect to α. If f is Riemann integrable with respect to α, then the *Riemann-Stieltjes integral* of f is defined by

$$\int_a^b f d\alpha = \int_{-a}^{b} f d\alpha = \int_{a}^{-b} f d\alpha$$

For the sake of conciseness, going forward, the reader should assume that all of the definitions provided above hold for the rest of the chapter and that the function f is always bounded on $[a, b]$. In addition, we refer to the Riemann-Stieltjes integral as simply the *Stieltjes integral*, and we say that a Riemann-Stieltjes integrabl function is simply *Stieltjes integrable.*

Note that the next several pages concern claims about the Stieltjes integral. However, recognize that Riemann integral is simply a Stietljes integral $\int_a^b f d\alpha$ with $\alpha = x$, so the claims apply to the Riemann integral as well.

Definition 1.4 (Fineness)**.** We say a partition P of $[a, b]$ is *finer* than a partition Q of $[a, b]$ if $P \subset Q$.

Remark. While we omit the proof, it follows from construction that for any partition P that is finer than partition Q, then $L(P, f, \alpha) \geq L(Q, f, \alpha)$, and $U(P, f, \alpha) \leq U(Q, f, \alpha)$.

Theorem 1.5. *f is Riemann integrable with respect to α i.e. $f \in \mathcal{R}(\alpha)$ if and only if for all $\varepsilon > 0$, there exists a partition P such that $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$.*

Proof. Let $\varepsilon > 0$. Suppose f is Riemann integrable with respect to α. Then,

$$\inf U(P, f, \alpha) = \int_{-a}^{b} f d\alpha = \int_{a}^{-b} f d\alpha = \sup L(P, f, \alpha) = A$$

for some $A \in \mathbb{R}$, so by construction, there exists partitions P_1, P_2 such that

$$U(P_1, f, \alpha) - A < \frac{\varepsilon}{2}$$

and

$$A - L(P_2, f, \alpha) < \frac{\varepsilon}{2}$$

Let $P = P_1 \cup P_2$. Then, since P is finer than P_1 and P_2,

$$U(P, f, \alpha) \leq U(P_1, f, \alpha) < A + \frac{\varepsilon}{2} < L(P_2, f, \alpha) + \varepsilon \leq L(P, f, \alpha) + \varepsilon$$
$$\Rightarrow U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

Now, suppose that for all $\varepsilon > 0$, there exists a partition P such that $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$. Since $L(P, f, \alpha) \leq \int_{-a}^{b} f d\alpha \leq \int_{a}^{-b} f d\alpha \leq U(P, f, \alpha)$, then it is implied that

$$0 \leq \int_{a}^{-b} f d\alpha - \int_{-a}^{b} f d\alpha < \varepsilon$$

Then, since it is assumed that the above is satisfied for all $\varepsilon > 0$, we conclude that $\int_{a}^{-b} f d\alpha = \int_{-a}^{b} f d\alpha$, and thus that $f \in \mathcal{R}(\alpha)$. $\square$

The following are some closely related properties of the Stietljes integral.

Theorem 1.6. Fix $\varepsilon > 0$.

1. If the inequality $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds for some partition P, then the inequality holds for any partition finer than P.
2. If the inequality $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds for partition $P = \{x_0, x_1, \ldots, x_n\}$, and if $s_i, t_i \in [x_{i-1}, x_i]$ for some $i \in \{1, \ldots, n\}$, then

$$\sum_{i=1}^{n} |f(s_i) - f(t_i)| \Delta\alpha_i < \varepsilon.$$

3. If $f \in \mathcal{R}(\alpha)$, and (2) holds, then

$$|\sum_{i=1}^{n} f(t_i) \Delta\alpha_i - \int_{a}^{b} f d\alpha| < \varepsilon.$$

Proof. 1. Suppose $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds for some fixed $\varepsilon > 0$. Then, for any partition P' that is finer than P, we have $U(P', f, \alpha) \leq U(P, f, \alpha)$ and $L(P', f, \alpha) \geq L(P, f, \alpha)$, so $U(P', f, \alpha) - L(P', f, \alpha) < \varepsilon$ as well.

2. Suppose $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds for partition $P = \{x_0, x_1, \ldots, x_n\}$. Then, for any $s_i, t_i \in [x_{i-1}, x_i]$ for each $i \in \{1, \ldots, n\}$,

$$\sum_{i=1}^{n} |f(s_i) - f(t_i)| \Delta\alpha_i \leq \sum_{i=1}^{n} | \sup_{[x_{i-1}, x_i]} f - \inf_{[x_{i-1}, x_i]} f | \Delta\alpha_i = U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

3. Suppose (2) holds. Then, since $L(P, f, \alpha) \leq \sum_{i=1}^{n} f(t_i) \Delta\alpha_i \leq U(P, f, \alpha)$, and $L(P, f, \alpha) \leq \int_{a}^{b} f d\alpha \leq U(P, f, \alpha)$,

$$|\sum_{i=1}^{n} f(t_i) \Delta\alpha_i - \int_{a}^{b} f d\alpha| \leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

$\square$

2. EXISTENCE OF THE RIEMANN-STIELTJES INTEGRAL

The following claims concern the criteria a function must satisfy to be Stieltjes integrable.

Theorem 1.7. If f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.

Proof. Suppose f is continuous on $[a, b]$ and let $\varepsilon > 0$. Notice that since $[a, b]$ is compact, f is uniformly continuous on $[a, b]$. Let $\gamma > 0$ so that

$$|\alpha(b) - \alpha(a)|\gamma < \varepsilon$$

Then, for some $\delta > 0$ and $x, y \in [a, b]$ such that $|x - y| < \delta$,

$$|f(x) - f(y)| < \gamma$$

Then, for any partition $P = \{x_0, x_1, \ldots, x_n\}$ such that $\Delta x_i < \delta$, we have $\sup_{[x_{i-1},x_i]} f - \inf_{[x_{i-1},x_i]} f \leq \gamma$ and thus,

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{i=1}^{n} \left(\sup_{[x_{i-1},x_i]} f - \inf_{[x_{i-1},x_i]} f \right) \Delta\alpha_i \leq |\alpha(b) - \alpha(a)|\gamma < \varepsilon$$

Then, by Theorem 2.5, $f \in \mathcal{R}(\alpha)$. □

Theorem 1.8. If f is monotonic on $[a, b]$ and α is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.

Proof. Let $\varepsilon > 0$. Since α is continuous on $[a, b]$, by the Intermediate Value Theorem, we can choose a partition P so that

$$\Delta\alpha_i = \frac{\alpha(b) - \alpha(a)}{n}$$

for each $i \in \{1, \ldots, n\}$.

Without loss of generality, suppose that f is monotonically increasing. Then $\sup_{[x_{i-1},x_i]} f = f(x_i)$ and $\inf_{[x_{i-1},x_i]} f = f(x_{i-1})$ for each $i \in \{1, \ldots, n\}$ and

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^{n} (f(x_i) - f(x_{i-1})) \\ &= \frac{\alpha(b) - \alpha(a)}{n} (f(b) - f(a)) < \varepsilon \end{aligned}$$

for large enough n. □

Theorem 1.9. If f has only finitely many points of discontinuity on $[a, b]$ and α is continuous at every point at which f is discontinuous, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.

Proof. Let $\varepsilon > 0$ and let E be the set of points of $[a, b]$ at which f is discontinuous. Since E is finite and α is continuous on E, we can cover E by finitely many disjoint intervals $[u_k, v_k] \subseteq [a, b]$ such that the sum of the differences $\alpha(v_k) - \alpha(u_k)$ is less than ε, and further, we can choose each u_k, v_k so that each point of E lies in the interior of some $[u_k, v_k]$.

Following, remove each open interval (u_k, v_k) from E, and denote the remaining subset of E as K. Notice that K is compact, and since f is continuous on K, f is also uniformly continuous on K. Thus, for some $\delta > 0$ and for $s, t \in K$ such that $|s - t| < \delta$, we have $|f(s) - f(t)| < \varepsilon$.

Following, define a partition $P = \{x_0, x_1, \ldots, x_n\}$ satisfying the following:

1. $u_k, v_k \in P$.

2. No point of (u_k, v_k) is in P.

3. If x_{i-1} is not u_k, then $\Delta x_i < \delta$.

Let $M \coloneqq \sup f(x)$. Then, notice that $\sup_{[x_{i-1},x_i]} f - \inf_{[x_{i-1},x_i]} f \leq 2M$ for all $i \in \{1, \ldots, n\}$, and $\sup_{[x_{i-1},x_i]} f - \inf_{[x_{i-1},x_i]} f \leq \varepsilon$ for $i \in \{1, \ldots, n\}$ such that $x_{i-1} \neq u_k$ for some k. Then,

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{i=1}^{n} \left(\sup_{[x_{i-1},x_i]} f - \inf_{[x_{i-1},x_i]} f \right) \Delta\alpha_i \leq (\alpha(b) - \alpha(a))\varepsilon + 2M\varepsilon$$

Notice that the term $(\alpha(b) - \alpha(a))\varepsilon$ accounts for $i \in \{1, \ldots, n\}$ such that $u_k \neq x_{i-1}$, and the term $2M\varepsilon$ accounts for $i \in \{1, \ldots, n\}$ such that $u_k = x_{i-1}$. Since ε was arbitrary, by Theorem 2.5, we conclude that $f \in \mathcal{R}(\alpha)$. □

Theorem 1.10. Let $f \in \mathcal{R}(\alpha)$ on $[a, b]$, and let $m, M \in \mathbb{R}$ be lower and upper bounds of f on $[a, b]$, respectively. If a real valued function g is continuous on $[m, M]$, then $h(x) \coloneqq (g \circ f)(x)$ defined on $[a, b]$ is Riemann integrable with respect to α on $[a, b]$ i.e. $h \in \mathcal{R}(\alpha)$ on $[a, b]$.

Proof. Let $\varepsilon > 0$. Since $[m, M]$ is compact and g is continuous on $[m, M]$, g is uniformly continuous on $[m, M]$. Thus, for some $\delta > 0$ such that $\delta < \varepsilon$ and $s, t \in [m, M]$ such that $|s - t| < \delta$, $|g(s) - g(t)| < \varepsilon$.

Then, since $f \in \mathcal{R}(\alpha)$, there exists a partition $P = \{x_0, x_1, \ldots, x_n\}$ such that $U(P, f, \alpha) - L(P, f, \alpha) < \delta^2$. Recall the definitions $M_i = \sup_{[x_{i-1}, x_i]} f$ and $m_i = \inf_{[x_{i-1}, x_i]} f$ for each $i \in \{1, \ldots, n\}$. Similarly, define $M_i' \coloneqq \sup_{[x_{i-1}, x_i]} h$ and $m_i' \coloneqq \inf_{[x_{i-1}, x_i]} h$ for each $i \in \{1, \ldots, n\}$.

Let $\{1, \ldots, n\}$ be the disjoint unions of two subsets $\mathcal{I}$ and $\mathcal{J}$, where $\mathcal{I}$ contains all indices i such that $M_i - m_i < \delta$, and $\mathcal{J}$ contains all indices j such that $M_j - m_j \geq \delta$. Then, $M_i' - m_i' < \varepsilon$ for all $i \in \mathcal{I}$ and $M_j' - m_j' \geq \varepsilon$ for all $j \in \mathcal{J}$.

Let $K \coloneqq \sup |g(x)|$ taken over $[m, M]$. Then, $M_j' - m_j' \leq 2K$, and we have

$$\delta \sum_{j \in \mathcal{J}} \Delta\alpha_j \leq \sum_{j \in \mathcal{J}} (M_j - m_j)\Delta\alpha_j \leq \delta^2,$$

so $\sum_{j \in \mathcal{J}} \Delta\alpha_j < \delta$. Then, recalling that we chose δ so $\delta < \varepsilon$,

$$\begin{aligned} U(P, h, \alpha) - L(P, h, \alpha) &= \sum_{i \in \mathcal{I}} (M_i' - m_i')\Delta\alpha_i + \sum_{j \in \mathcal{J}} (M_j' - m_j')\Delta\alpha_j \\ &\leq (\alpha(b) - \alpha(a))\varepsilon + 2K\delta \\ &\leq (\alpha(b) - \alpha(a))\varepsilon + 2K\varepsilon \end{aligned}$$

Since ε was arbitrary, by Theorem 2.5, we conclude that $h \in \mathcal{R}(\alpha)$. □

Theorem 1.11. Let α be differentiable on (a, b), and $\alpha' \in \mathcal{R}(x)$. Then, $f \in \mathcal{R}(\alpha)$ on $[a, b]$ if and only if $f\alpha' \in \mathcal{R}(x)$ on $[a, b]$ and

$$\int_a^b f d\alpha = \int_a^b f\alpha' dx.$$

Proof. WILL FINISH IN CLASS □

Theorem 1.12. If $\{f_n\}_{n \geq 1}$ is a family of functions in $\mathcal{R}$ on $[a, b]$ and $\{f_n\}_{n \geq 1}$ converges uniformly to some function f that is bounded on $[a, b]$, then $f \in \mathcal{R}$ and

$$\int_a^b f_n(x)dx \to \int_a^b f(x)dx.$$

3. PROPERTIES OF THE RIEMANN-STIELTJES INTEGRAL

Theorem 1.13.

1. If $f_1, f_2 \in \mathcal{R}(\alpha)$, then $f_1 + f_2 \in \mathcal{R}(a)$ and

$$\int_a^b (f_1 + f_2) d\alpha = \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha.$$

2. If $f \in \mathcal{R}(\alpha)$, then $cf \in \mathcal{R}(\alpha)$ for any $c \in \mathbb{R}$, and

$$\int_a^b cf d\alpha = c \int_a^b f d\alpha.$$

3. If $f_1, f_2 \in \mathcal{R}(\alpha)$ such that $f_1(x) \leq f_2(x)$ for all $x \in [a,b]$, then
$$\int_a^b f_1 d\alpha \leq \int_a^b f_2 d\alpha.$$
4. If $f \in \mathcal{R}(\alpha)$ on $[a,b]$, then for any $c \in (a,b)$, $f \in \mathcal{R}(\alpha)$ on $[a,c]$ and $f \in \mathcal{R}(\alpha)$ on $[c,b]$, and
$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha.$$
5. If $f \in \mathcal{R}(\alpha)$ on $[a,b]$ and f is bounded above by M on $[a,b]$, then
$$\int_a^b f d\alpha \leq M(\alpha(b) - \alpha(a)).$$
6. If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$ for distinct monotonically increasing functions α_1 and α_2 on $[a,b]$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and
$$\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2.$$
7. If $f \in \mathcal{R}(\alpha)$, then for any $c > 0$, $f \in \mathcal{R}(c\alpha)$ and
$$\int_a^b f d(c\alpha) = c \int_a^b f d\alpha.$$

Proof. All claims above follow almost exactly from definition or have proofs very similar to previous theorems in this chapter, so we omit their proofs. □

Theorem 1.14. If $f, g \in \mathcal{R}(\alpha)$ on $[a,b]$, then

1. $fg \in \mathcal{R}(\alpha)$.
2. $|f| \in \mathcal{R}(\alpha)$ and
$$|\int_a^b f d\alpha| \leq \int_a^b |f| d\alpha.$$

Proof. 1. Notice that $4fg = (f+g)^2 - (f-g)^2$. Since $f+g, f-g \in \mathcal{R}(\alpha)$, and by Theorem 1.10, $(f+g)^2, (f-g)^2 \in \mathcal{R}(\alpha)$ as well, $4fg \in \mathcal{R}(\alpha)$. Thus, taking $c = \frac{1}{4}$,
$$c(4fg) = fg \in \mathcal{R}(\alpha).$$

2. By Theorem 1.10, $|f| \in \mathcal{R}(\alpha)$. Then, we can choose $c = \pm 1$ so that
$$|\int_a^b f d\alpha| = c \int_a^b f d\alpha = \int_a^b c f d\alpha \leq \int_a^b |f| d\alpha.$$
□

Definition 1.15 (Step Function)**.** We define the step function I as the real valued function
$$I(x) := \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Theorem 1.16. Let $s \in (a,b)$, f be a real-valued function bounded on $[a,b]$, f be continuous at s, and $\alpha(x) = I(x-s)$. Then,
$$\int_a^b f d\alpha = f(s).$$

Proof. Consider the partition $P = \{x_0, x_1, x_2, x_3\}$, where $a = x_0 < x_1 = s < x_2 < x_3 = b$. Then, $U(P,f,\alpha) = M_2$ and $L(P,f,\alpha) = m_2$, and since f is continuous at s, as $x_2 \to s$, $M_2 \to f(s)$ and $m_2 \to f(s)$. Thus, $\int_a^b f d\alpha = f(s)$. □

4. EXISTENCE OF THE RIEMANN INTEGRAL

The following claims concern the criteria a function must satisfy to be Riemann integrable (but not necessarily Stieltjes integrable).

Definition 1.17 (Mesh)**.** Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. The *mesh* of P is defined

$$\operatorname{mesh}(P) \coloneqq \sup_{\{1,\dots,n\}} \Delta x_i.$$

Theorem 1.18. $f \in \mathcal{R}$ if and only if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if a partition P of $[a, b]$ has $\operatorname{mesh}(P) < \delta$, then $U(P, f, x) - L(P, f, x) < \varepsilon$.

Proof. Suppose that for all $\varepsilon > 0$, there exists $\delta > 0$ such that if a partition P of $[a, b]$ with $\operatorname{mesh}(P) < \delta$, then $U(P, f) - L(P, f) < \varepsilon$. By Theorem 1.5, $f \in \mathcal{R}$.

Now, suppose that $f \in \mathcal{R}$, and let $\varepsilon > 0$. By Theorem 1.5, there exists a partition $P_0 = \{x_0, x_1, \dots, x_n\}$ such that $U(P_0, f) - L(P_0, f) < \frac{\varepsilon}{2}$. Let $\operatorname{mesh}(P_0) = \delta_0$, Following, choose $\delta > 0$ such that for $y_k \coloneqq y_1 + k\delta$, $[y_{k-1}, y_k]$ contains at most one x_i for each k, and let $P = \{y_0, y_1, \dots, y_m\}$ denote this partition. Then,

$$\begin{aligned} U(P, f) - U(P_0, f) &\leq U(P, f) - U(P \cup P_0, f) \\ &= \sum_{k=1}^{m} \sup_{[y_{k-1}, y_k]} |f|(y_k - y_{k-1}) - \sum_{k=1}^{m} \Big(\sup_{[y_{k-1}, x_k]} |f|(x_k - y_{k-1}) + \sup_{[x_k, y_k]} |f|(y_k - x_k) \Big) \\ &= \sum_{k=1}^{m} \Big(\sup_{[y_{k-1}, y_k]} |f| - \sup_{[y_{k-1}, x_k]} |f| \Big)(x_k - y_{k-1}) + \Big(\sup_{[y_{k-1}, y_k]} |f| - \sup_{[x_k, y_k]} |f| \Big)(y_k - x_k) \\ &< \sum_{k=1}^{m} \Big(\sup_{[y_{k-1}, y_k]} |f| - \sup_{[y_{k-1}, x_k]} |f| \Big)\delta + \Big(\sup_{[y_{k-1}, y_k]} |f| - \sup_{[x_k, y_k]} |f| \Big)\delta \\ &< 2M\delta \end{aligned}$$

Similarly, we see that $L(P, f) - L(P_0, f) > -2M\delta$. Then, for $\delta = \frac{\varepsilon}{8M}$, we have

$$\begin{aligned} &\begin{cases} U(P, f) - U(P_0, f) < \frac{\varepsilon}{4} \\ L(P_0, f) - L(P, f) < \frac{\varepsilon}{4} \end{cases} \\ &\Rightarrow U(P, f) - \frac{\varepsilon}{2} - L(P, f) < U(P, f) - U(P_0, f) + L(P_0, f) - L(P, f) < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2} \\ &\Rightarrow U(P, f) - L(P, f) < \varepsilon \end{aligned}$$

so $\delta = \min\{\delta_0, \frac{\varepsilon}{8M}\}$. □

Definition 1.19 (Riemann Sum)**.** Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. A *Riemann sum* $S(P, f)$ is

$$S(P, f) \coloneqq \sum_{i=1}^{n} f(t_i)\Delta x_i$$

where $t_i \in [x_{i-1}, x_i]$ for each $i \in \{1, \dots, n\}$.

Theorem 1.20. $f \in \mathcal{R}$ if and only if there exists $r \in \mathbb{R}$ such that for all $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|S(P, f) - r| < \varepsilon$$

for any Riemann sum $S(P, f)$ with partition P satisfying $\operatorname{mesh}(P) < \delta$.

Proof. Suppose $f \in \mathcal{R}$ and let $\varepsilon > 0$. Then, by Theorem 1.17, there exists $\delta > 0$ such that if $\text{mesh}(P) < \delta$, then $U(P,f) - L(P,f) < \varepsilon$. Then, since $L(P,f) \leq \int_a^b f dx \leq U(P,f)$ and $L(P,f) \leq S(P,f) \leq U(P,f)$,

$$\begin{cases} U(P,f) - \int_a^b f dx < \varepsilon \\ L(P,f) - \int_a^b f dx > -\varepsilon \end{cases} \Rightarrow \begin{cases} S(P,f) - \int_a^b f dx < \varepsilon \\ S(P,f) - \int_a^b f dx > -\varepsilon \end{cases} \Rightarrow |S(P,f) - \int_a^b f dx| < \varepsilon$$

Thus, $|S(P,f) - r| < \varepsilon$ is satisfied when $r = \int_a^b f dx$.

Suppose that for all $\varepsilon > 0$, there exists $\delta > 0$ such that $|S(P,f) - r| < \varepsilon$ for any Riemann sum $S(P,f)$ with partition $P = \{x_0, x_1, \ldots, x_n\}$ satisfying $\text{mesh}(P) < \delta$. Then, for each $i \in \{1, \ldots, n\}$, there exists $s_i, t_i \in [x_{i-1}, x_i]$ such that

$$\begin{aligned} f(s_i) &> \sup_{[x_{i-1},x_i]} f + \frac{\varepsilon}{2(b-a)} \\ f(t_i) &< \inf_{[x_{i-1},x_i]} f - \frac{\varepsilon}{2(b-a)} \end{aligned}$$

Then, notice that

$$\begin{aligned} S_1(P,f) &\coloneqq \sum_{i=1}^{n} f(s_i)\Delta x_i > (U(P,f) - \frac{\varepsilon}{2(b-a)}) \sum_{i=1}^{n} \Delta x_i = U(P,f) - \frac{\varepsilon}{2} \\ S_2(P,f) &\coloneqq \sum_{i=1}^{n} f(t_i)\Delta x_i < (L(P,f) + \frac{\varepsilon}{2(b-a)}) \sum_{i=1}^{n} \Delta x_i = L(P,f) + \frac{\varepsilon}{2} \end{aligned}$$

so

$$\begin{aligned} &\begin{cases} U(P,f) - \frac{\varepsilon}{2} < S_2(P,f) < r + \frac{\varepsilon}{2} \\ r - \frac{\varepsilon}{2} < S_1(P,f) < L(P,f) + \frac{\varepsilon}{2} \end{cases} \\ &\Rightarrow U(P,f) - L(P,f) = U(P,f) - \frac{\varepsilon}{2} - (L(P,f) + \frac{\varepsilon}{2}) < r + \frac{\varepsilon}{2} - (r - \frac{\varepsilon}{2}) = \varepsilon \end{aligned}$$

Thus, by Theorem 1.5, $f \in \mathcal{R}$. $\square$